Unit-5 (Data mining task, Technique and Tools)

BScCSIT 8th Semester

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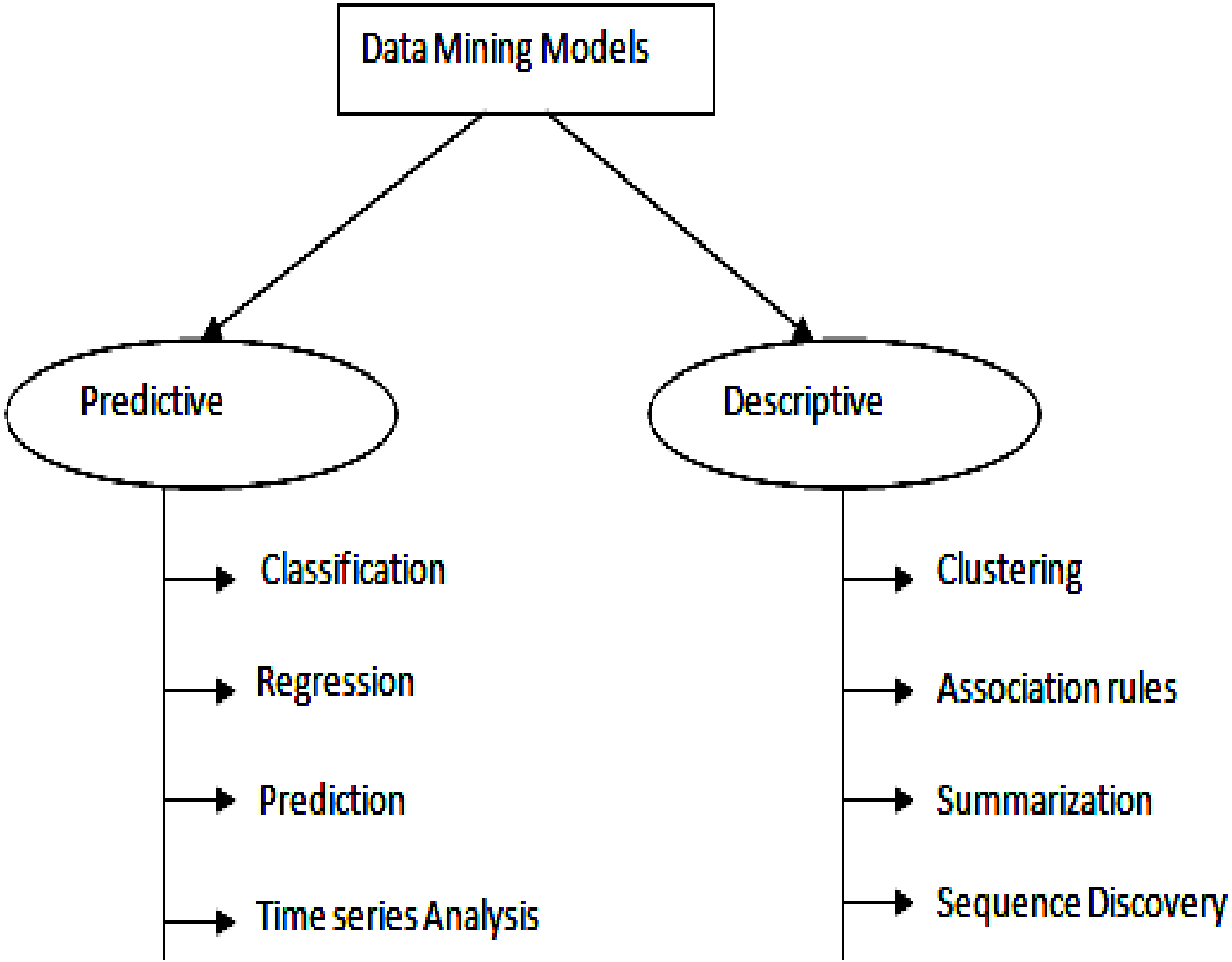
Models or Task of data mining

- Data mining evolves many different algorithms to accomplish different tasks. All of these algorithm attempt to fit a model to the data. The algorithm attempt to fit a model to the data.
- The algorithm examines the data and determines a model that is closest to the characteristics of the data being examined. There are two types of data mining models.
  - **Predictive Data models:**
  - **Descriptive data models:**


Data Mining Models
Predictive
Descriptive
Classification
Regression
Prediction
Time series Analysis
Clustering
Association rules
Summarization
Sequence Discovery

Predictive Data models

- It makes a prediction about values of data using known results found from different data.
- **Some of tasks of this model are:**
- classification,
- regression,
- prediction, and
- time-series analysis.

Descriptive data models

- It identifies the patterns or relationships in data. Unlike the predictive model, a descriptive model serves as a way to explore the properties of the data examined not to predict new properties.
- Some of the tasks of this model are:
- clustering,
- summarization,
- sequence discovery and
- association rules.

Data mining tasks and methodologies

- **Classification**
- **Regression**
- **Prediction**
- **Sequence Discovery**
- **Summarization**
- **Clustering**
- **Association rules**
- **Time series analysis**

Classification

- Classification is learning rules that can be applied to new data and will typically include following steps: preprocessing of data, designing modeling, learning/feature selection and validation /evaluation.
- Classification predicts categorical continuous valued functions.
- For example, we can make classification model to categorize bank loan application as either safe or risky.
- Classification is the derivation of model which determines the class of an object based on its attributes.
- A set of object is given as training set in which every object is represented by vector of attributes along with its class. By analyzing the relationship between attributes and class of the objects in the training set, classification model can be constructed.
- Classification model can be used to classify future objects and develop a better understanding of the classes of the objects in the data base.
- For example, from the set of loan borrowers (Name, Age, and Income) who serve as training set, a classification model can be built, which concludes bank loan application as either safe or risky. (If age = Youth then Loan decision = risky).

Regression

- Regression is finding function with minimal error to model data.
- It is statistical methodology that is most often used for numeric prediction.
- Regression analysis is widely used for prediction and forecasting, where its use has substantial overlap with the field of machine learning.
- Regression analysis is also used to understand which among the independent variables are related to the dependent variable, and to explore the forms of these relationships.
- In restricted circumstances, regression analysis can be used to infer causal relationships between the independent and dependent variables.

Prediction

- This mining function predicts the possible values of soe missing data or the value distribution of certain attributes in a set of objects.
- It involves the finding of the set of attributes relevant to the attribute of interest (e.g. by some statistical analysis) and predicting the value distribution based on the set of data similar to the selected objects.
- For example, an employee's potential salary can be predicted based on the salary distribution of similar employees in the company.
- Usually, regression analysis, generalized linear model, correlation analysis and decision trees are useful tools in quality prediction. Genetic algorithms and neural network models are also popularly used in prediction.

Sequence Discovery

- It is used to analyze and discover the sequential patterns in data.
- These patterns are based on the time sequence of actions.
- E.g. most people who purchase the CD player may be found to purchase CDs within one week.

Summarization

- Summarization is the generalization or abstraction of data.
- A set of relevant data is abstracted and summarized,
- resulting a smaller set which gives a general overview of data.
- For example, the long distance calls of customer can be summarized in to total minutes, total calls, total spending etc instead of detailed calls. Similarly the calls can be summarized in to local calls, STD calls, ISD calls etc.

Clustering

- Clustering is identifying similar groups from unstructured data.
- Clustering is the task of grouping a set of objects in a such a way that object in same group are more similar to each other than to those in other groups.
- Once the clusters are decided, the objects are labeled their corresponding clusters, and common features of the objects in cluster are summarized to form a class description.
- For example, a bank may cluster its customer in to several groups based on the similarities of their income, age, sex, residence etc, and the command characteristics of the customers in a group can be used to describe that group of customers. This will help the bank to understand its customers better and provide customized services.

Association rules

- Association is looking for relationship between variables or objects.
- It aims to extract interesting association, correlations or casual structures among the objects i.e. the appearance of another set of objects in.
- The association rules can be useful for marketing, commodity management, advertising etc.
- Association rule learning is a popular and well researched method for discovering interesting relations between variables in large databases.
- It is intended to identify strong rules discovered in databases using different measures of interestingness and based on the concept of strong rules presented in , introduced association rules for discovering regularities between products in large-scale transaction data recorded by point-of-sale (POS) systems in supermarkets.

Association rules

- For example, the rule {Onions, potatoes} {burger} found in the sales data of a supermarket would indicate that if a customer buys onions and potatoes together, he or she is likely to also buy hamburger meat. Such information can be used as the basis for decisions about marketing activities such as, e.g., promotional pricing or product placements.
- In addition to the above example from market basket analysis association rules are employed today in many application areas including Web usage mining, intrusion detection, Continuous production, and bioinformatics.

Time series analysis

- Time-series analysis is to analyze large set of time-data to find certain regularities and interesting characteristics, including search for similar sequences or subsequences, and mining sequential patterns, periodicities, trends and deviations.
- For example, one may predict the trend of the stock values for a company based on its stock history, business situation, competitor's performance, and current market.

Data Mining techniques

- Data mining techniques provides you with a way to use various data mining tasks.
- Data mining adopt its technique from many research areas, including statics, machine learning, database systems, rough sets, visualization and neural networks.
- Some of the data mining techniques are:-
  - **Statistical Approach**
  - **Machine learning**
  - **Decision tree**
  - **Visualization techniques**
  - **Association rules**
  - **Neural networks**
  - **Genetic algorithms**

Statistical Approach

- Statistics is a branch of mathematics, which deals with the collection and analysis of numeric data by using various methods and techniques like:-
- Point estimation
- Data summarization
- Bayesian theorem
- Correlation
- Chi-square test (i.e. hypothesis test)
- Regression
- Mean, median, mode etc.

Statistical Approach

- Statistical models are built from a set of training data.
- Many statistical tools have been used for data mining including, Bayesian network, correlation analysis, regression analysis and cluster analysis.
- For example simple Bayesian network for traffic jam problem is given in figure 2.
- In the Bayesian network nodes represents states or variable while edges represents dependencies between nodes.
- From figure we can see that rush hour, bad weather or accident affect the traffic which in turn causes traffic jam.

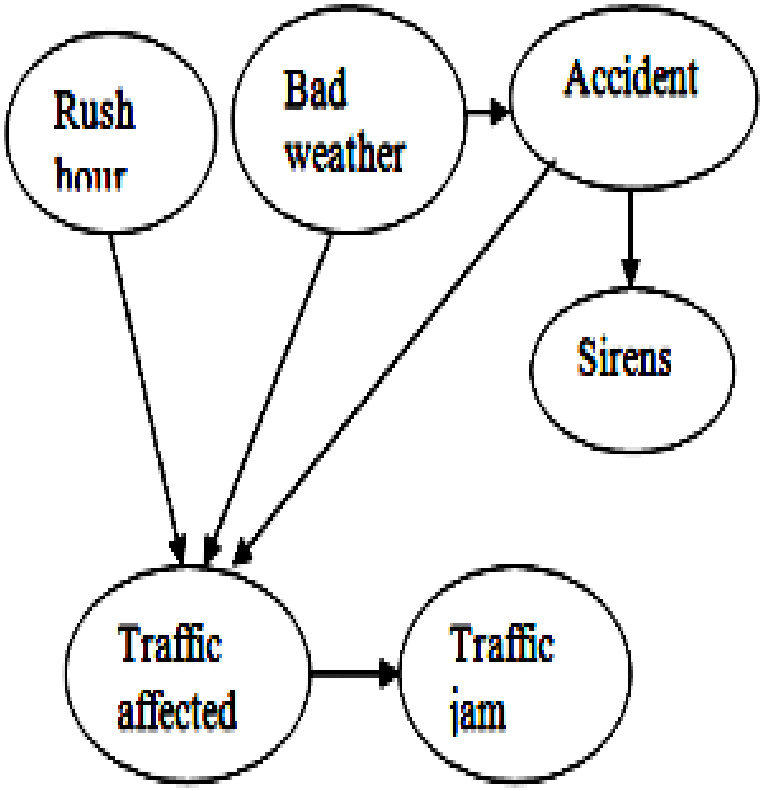


Figure 2. Example of an unacceptable low-resolution image

Machine learning

- It is a sub discipline of computer science that deals with the design and implementation of learning algorithm.
- It is the process of generating a computer system that is capable independently acquiring the data and integrating that data to generate useful knowledge.
- The concept of machine learning is to make a computer system behave and act as a human being who learns from experience, analysis the observations made, such a system is developed that continuously self-improves and provides increased efficiency and effectiveness.
- The use of machine learning in data mining is that machine learning enables you to discover new and interesting structures and formats about a set of data that are previously unknown.
- The most common machine learning methods used for data mining include conceptual learning, inductive concept learning and decision tree induction.

Decision tree

Definition:

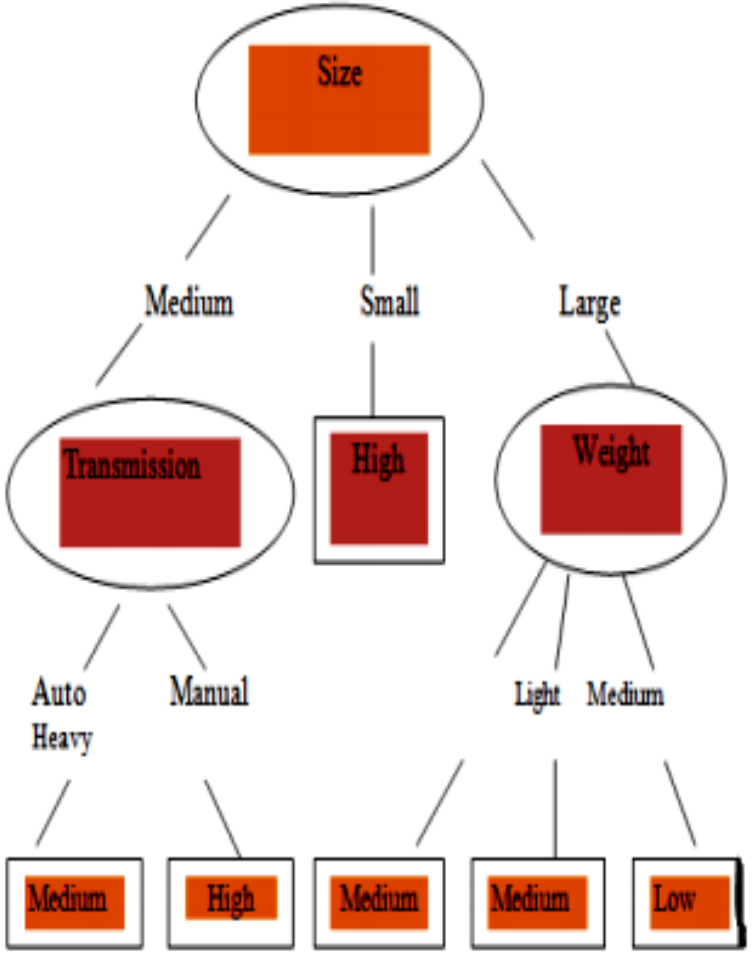
- A decision tree (DT) is a tree where the root and each internal node are labeled with a question.
  The arcs emanating from each node represents each possible answer to the associated question. Each leaf node represents a prediction of a solution to the problem under consideration.
- A DT model is computational models that have 3-parts:-
  - A decision tree as defined above
  - An algorithm to create the tree
  - An algorithm that applies the tree to data and solves the problem under consideration.

Decision tree

- A decision tree is a tree where each non-terminal node represents a test or decision on the considered data item.
- Choice of a certain branch depends upon the outcome of the test.
- To classify a particular data item, we start at the root node and follow the assertions down until we reach a terminal node (or leaf).
- A decision is made when a terminal node is approached.
- Decision trees can also be taken as a special form of a rule set, characterized by their hierarchical group of rules.

DT Example

- A simple decision tree is given in Figure. It determines the car's mileage from its size, transmission type and weight. The leaf nodes are in square boxes.



Visualization techniques

- Visualization refers to the visual representation of the data. This allows the users to summarize, extract and grasp the more complex results than more mathematical or text type description of the results e.g.
- **Graphical**: Pie chart, histograms, bar diagram, lines etc.
- **Geometric**: Box plot, scatter diagrams etc
- **Icon based**: Images, figures, colors etc
- **Pixel based**: Colored pixels
- **Hybrid**: Use of all above method in hybrid approach

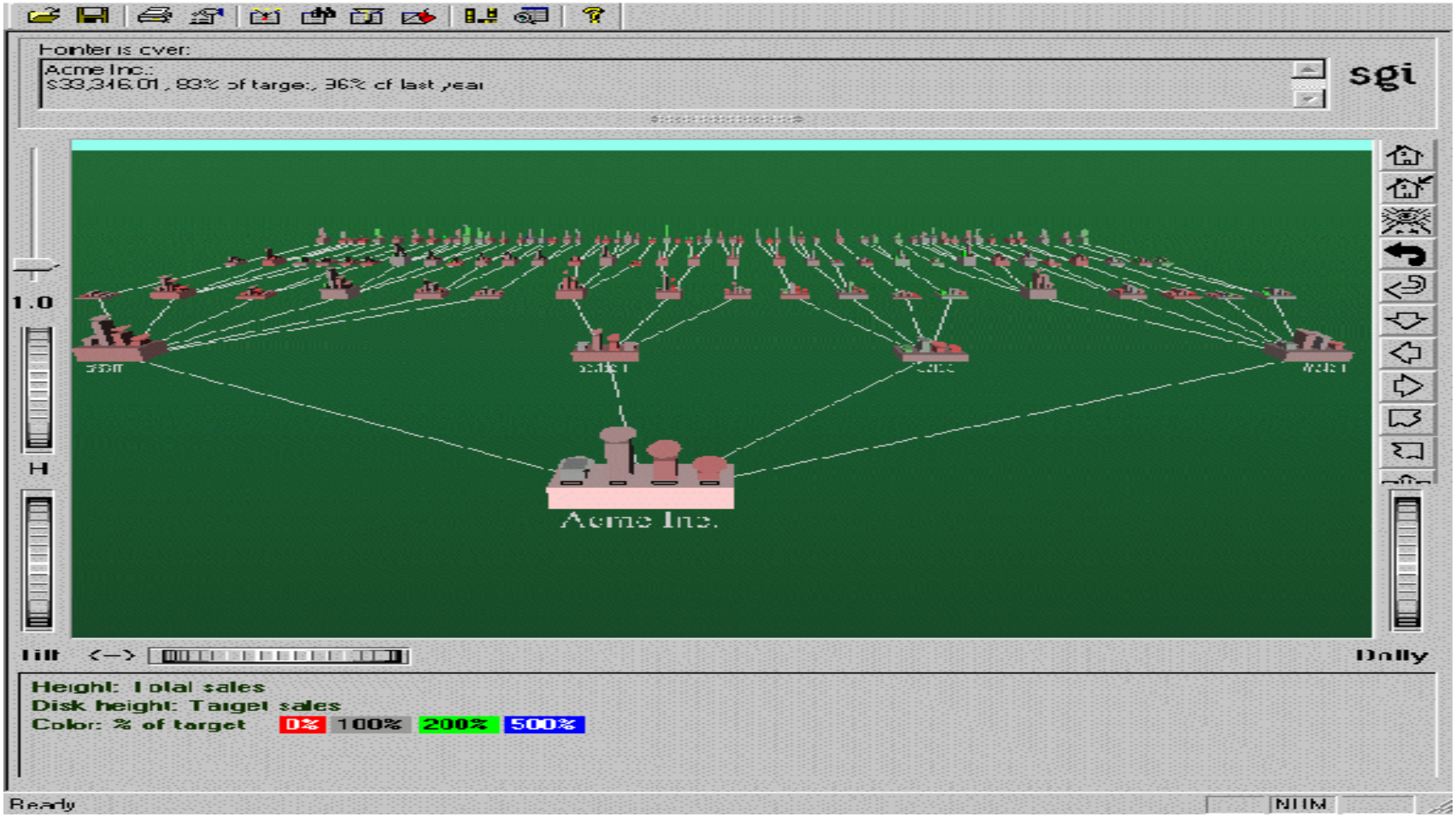
Visualization

- Purpose of Visualization
  - Gain insight into an information space by mapping data onto graphical primitives
  - Provide qualitative overview of large data sets
  - Search for patterns, trends, structure, irregularities, relationships among data.
  - Help find interesting regions and suitable parameters for further quantitative analysis.
  - Provide a visual proof of computer representations derived

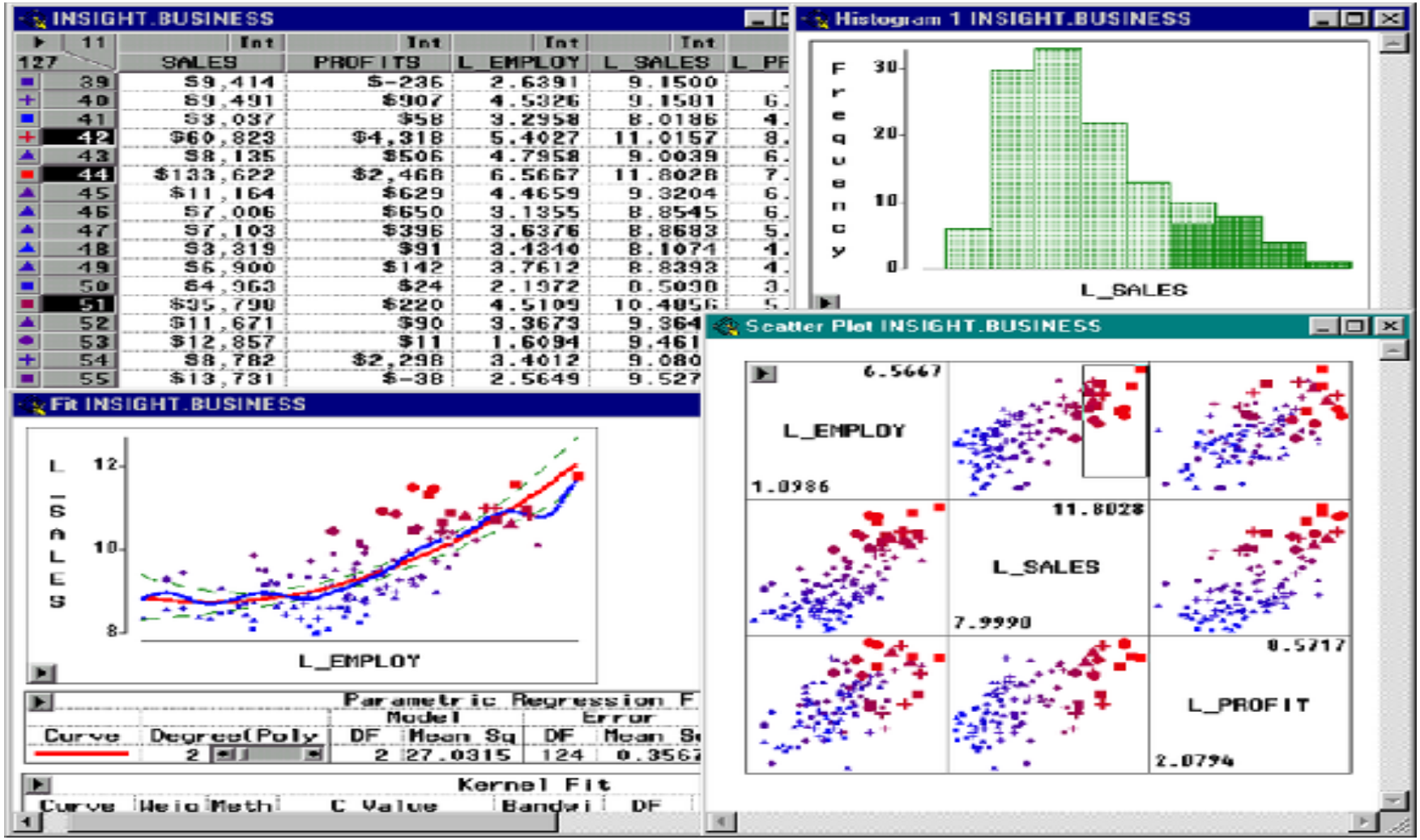
Examples

- Scatter plots and box plots (obtained from descriptive data mining)
- Decision trees
- Association rules
- Clusters
- Outliers
- Generalized rules

Visualization of a Decision Tree in SGI/MineSet 3.0



Visualization of Data Mining Results in SAS Enterprise Miner: Scatter Plots



Neural networks

- Neural network, often referred to as artificial neural network (ANN) are modeled after the working of human brain.
- The NN is actually an information processing system that consists of a graph representing the processing system as well as various algorithms that access that graph.
- As with the human brain, the NN consists of many connected processing elements. The NN then is structured as a directed graph with many nodes (processing elements) and arcs (interconnection) between them.
- The nodes in the graph are like individual neurons while arc are their interconnections.
- The NN can be viewed as a directed graph with source (input), sink (output) and internal (hidden) nodes.
- The input nodes exist in an input layer, while the output node exists in an output layer.
- The hidden nodes exist over one or more hidden layers to perform a data mining task, a tuple is input through the input nodes and the output nodes determines what the prediction is?

A neural network model is a computational model consisting of 3 parts:-

a. Neural network graph that defines the data structure of the neural network
b. Learning algorithm that indicates how learning take place.
c. Recall technique that determines how information is obtained from the network.

Some of the forms of neural network are:-

- Perceptrons
- Back propagation network
- NN supervised learning
- Radial basis function network
- Kohonen-self organizing map

a) Perceptrons:

A Perceptron is a single neuron with multiple inputs and outputs. These are simple feed forward neural network. E.g.

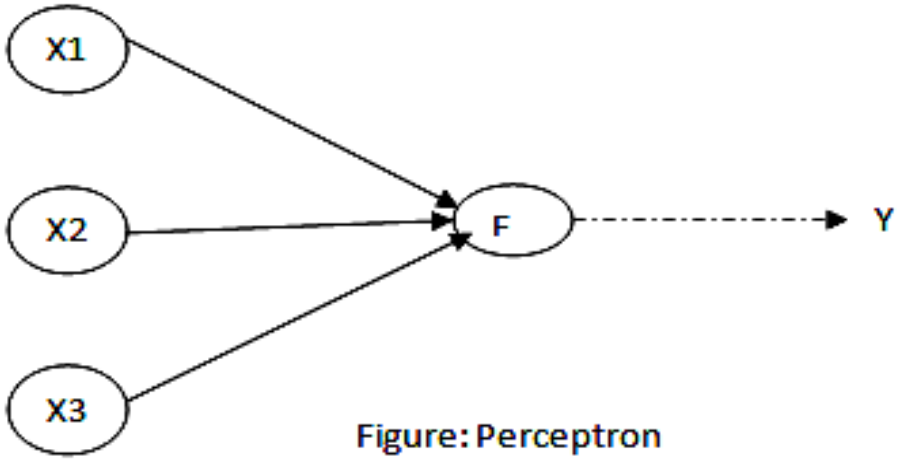
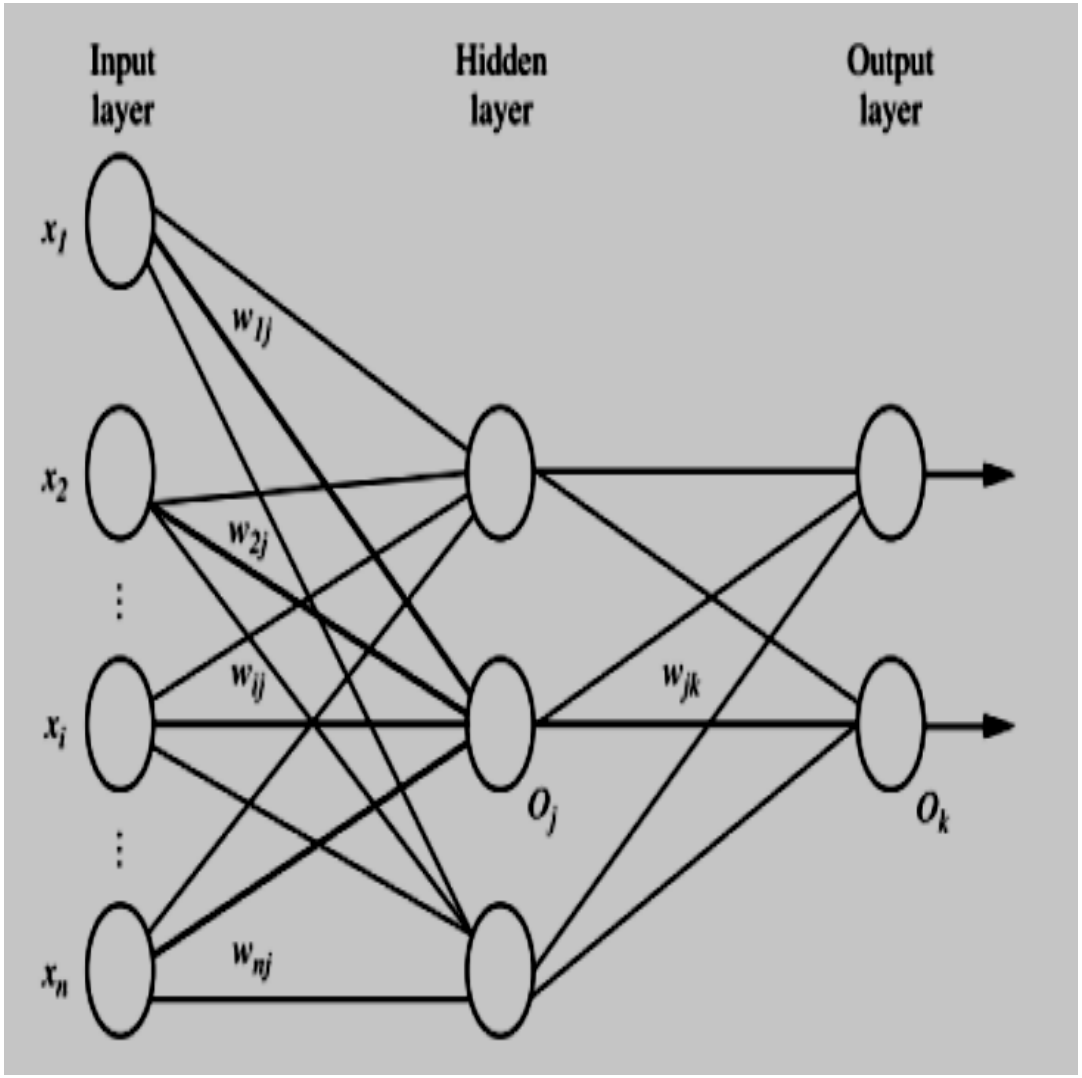


Figure: Perceptron

Back propagation network

- The back propagation algorithm performs learning on a multilayer feed-forward neural network.
- It iteratively learns a set of weights for prediction of the class label of tuples.
- A multilayer feed-forward neural network consists of an input layer, one or more hidden layers, and an output layer.
- An example of a multilayer feed-forward network is shown in figure.



Kohonen self-organizing map

- In 1981 Tuevo Kohonen demonstrated a completely different version of neural network called Kohonen self-organizing map.
- A Kohonen self-organizing map is a collection of neurons or units called its neighbors. It is a two dimensional map.
- Each node/unit contains a factor that is related to the space. In its initial setting, the map has a random vectors assigned to each unit.
- Only at training stage, these vectors are adjusted to give a better coverage of the space.

Advantages of Neural Network

- prediction accuracy is generally high
- robust, works when training examples contain errors
- output may be discrete, real-valued, or a vector of several discrete or real-valued attributes
- fast evaluation of the learned target function
- High tolerance to noisy data
- Ability to classify untrained patterns
- Well-suited for continuous-valued inputs and outputs
- Successful on a wide array of real-world data
- Algorithms are inherently parallel
- Techniques have recently been developed for the extraction of rules from trained neural networks

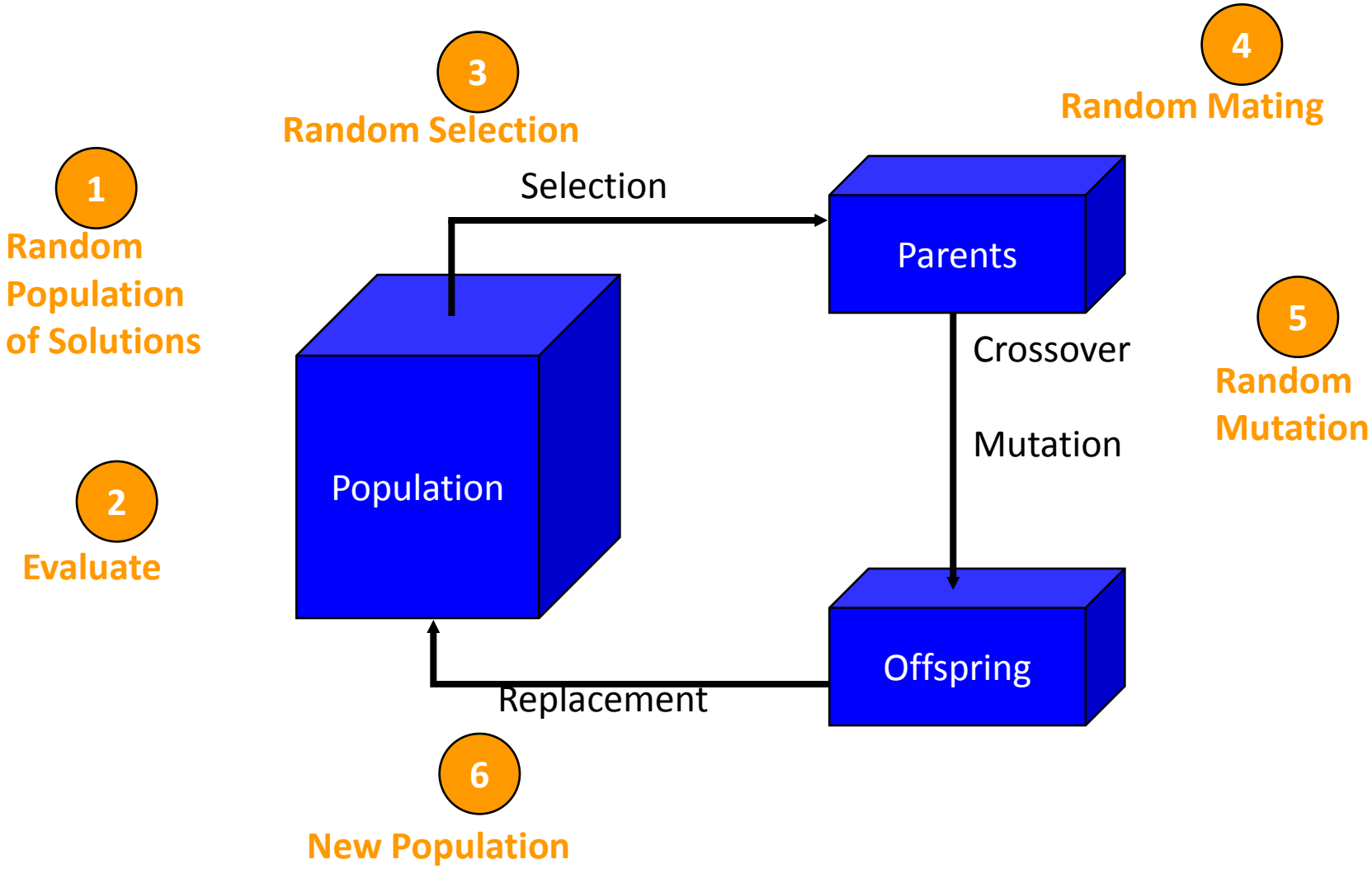
Disadvantages of Neural Network

- long training time
- difficult to understand the learned function (weights)
- not easy to incorporate domain knowledge
- Require a number of parameters typically best determined empirically, e.g., the network topology or ``structure."
- *Poor interpretability:* Difficult to interpret the symbolic meaning behind the learned weights and of ``hidden units" in the network

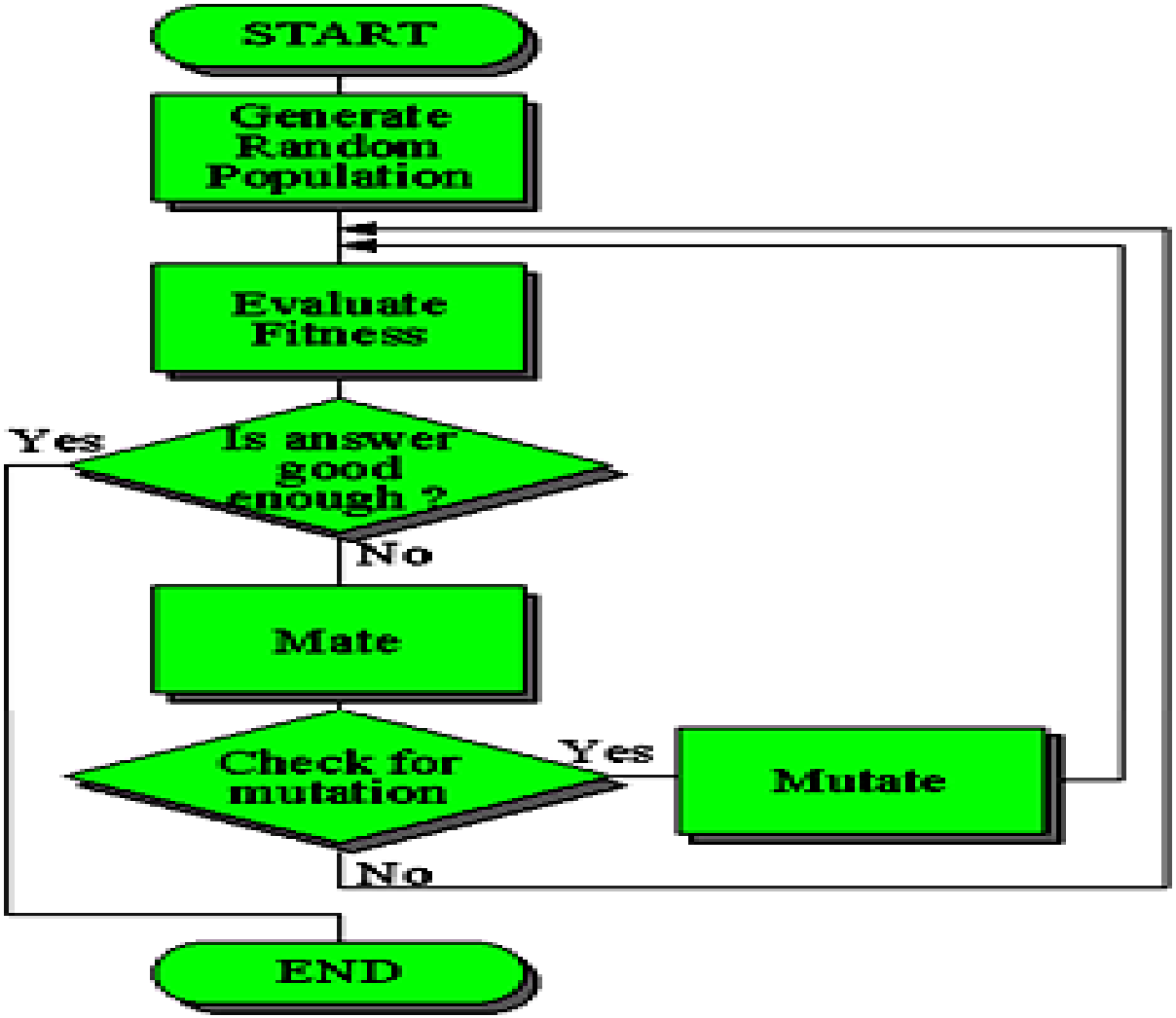
Genetic algorithms

- Genetic algorithms are a method of breeding computer solutions to optimization problems by simulated evolution.
- The processes based on crossover and mutation repeatedly applied to a population of binary strings.
- Time after time, the better fit individuals and average individuals are created, and a good solution to the problem is found.
- Genetic algorithms usually are used to predict and used to replace the missing attributes.
- It means when there are some attributes missed, we can analyses its specimens using genetic algorithms to get the possible value and replace the missing one.

General Approach



Flow chart


START
Generate Random Population
Evaluate Fitness
Is answer good enough ?
Yes
No
Mate
Check for mutation
Yes
Mutate
No
END

GA Applications

- **Numerical, Combinatorial Optimization**
- **System Modeling and Identification**
- **Planning and Control**
- **Engineering Design**
- **Data Mining**
- **Machine Learning**
- **Artificial Life**

Advantages and Disadvantage of GA

Advantages:

- No presumptions w.r.t. problem space
- Widely applicable
- Low development & application costs
- Easy to incorporate other methods
- Solutions are interpretable (unlike NN)
- Can be run interactively, accommodate user proposed solutions
- Provide many alternative solutions

Disadvantages

- No guarantee for optimal solution within finite time
- Weak theoretical basis
- May need parameter tuning
- Often computationally expensive, i.e. slow

Data Mining Tools

SPSS®

WEKA
The University of Waikato

ORACLE®

Microsoft®
SQL Server® 2012

WEKA

WEKA (Waikato Environment for Knowledge Analysis) is a popular suite of machine learning software written in Java, developed at the University of Waikato, New Zealand.

WEKA is free software available under the GNU General Public License.

Features:

- Written in JAVA
- Has graphical user interfaces
- Contains a collection of visualization tools and algorithms for data analysis and predictive modeling
- Supports standard data mining tasks like data preprocessing, clustering, classification, regression, visualization, and feature selection

Usage:

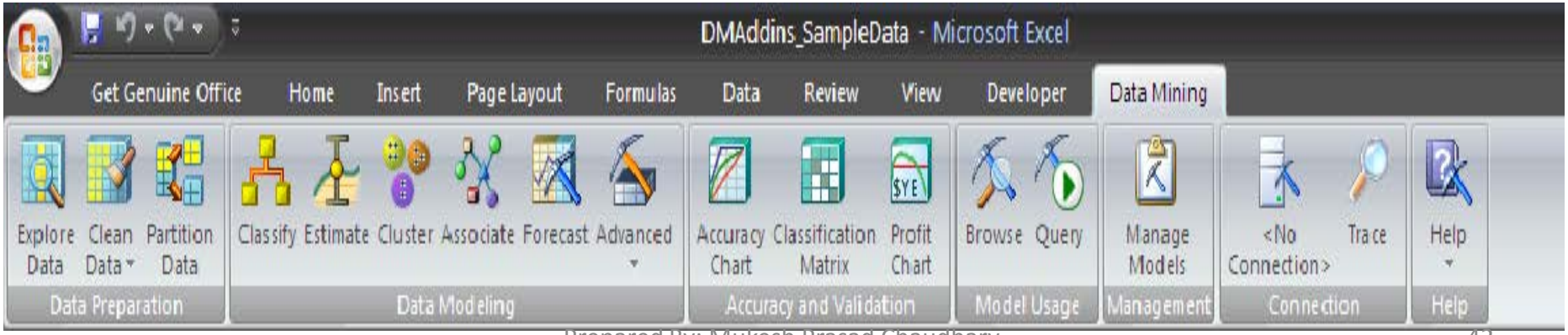
- Apply a learning method to a dataset & analyze the result
- Use a learned model to make predictions on new instances
- Apply different learners to a dataset & compare results

MS Excel

In order to bridge the gap between the common user and the complex data mining process, Microsoft has introduced a new and efficient data mining tool ,the Microsoft SQL Server 2005 Data Mining Add-Ins for Office 2007 putting data mining within the reach of every user or desktop.

The add-in can be downloaded from the DOWNLOAD LINK.

- The software pre-requisites for using the add-in are:
- Microsoft Office 2007 installed.
- Microsoft SQL Server 2005 or above installed.
  - Microsoft .NET 2.0 framework or higher (for SQL server 2008 only).
  - Microsoft PowerShell (for SQL server 2008 only)



Once the add-in is installed, you can see the DATA MINING tab in the EXCEL ribbon. The tab contains different options like:

Data Preparation

- Explore Data.
- Clean Data.
- Partition Data.

Data Modeling

- Classify.
- Estimate.
- Cluster.
- Associate.
- Forecast.
- Advanced.

Accuracy and Validation

- Accuracy Chart.
- Classification Matrix.
- Profit Chart.

Model Usage

- Browse.
- Query.

Management

Connection

- No Connection.
- Trace.

Help

Conclusion

The Microsoft SQL Server Data mining add-in for Microsoft Excel provides users with an easy to use interface that is capable of performing complex data mining tasks with ease. The add-in can be extremely useful for both, people who just want to get more out of their data and also for those interested in serious data mining.

Tanagra

- Tanagra is free data mining software for academic and research purposes.
- It offers several data mining methods like exploratory data analysis, statistical learning and machine learning.
- The **first purpose** of the Tanagra project is to give researchers and students easy-to-use data mining software.
- The **second purpose** of TANAGRA is to propose to researchers an architecture allowing them to easily add their own data mining methods, to compare their
- performances.
- The **third and last purpose** is that novice developers should take advantage of the free access to source code, to look how this sort of software was built, the problems to avoid, the main steps of the project, and which tools and code libraries to use for.
- In this way, Tanagra can be considered as a pedagogical tool for learning programming techniques as well

Orange

- **Orange** is a library of C++ core objects and routines that includes a large variety of standard and not-so-standard machine learning and data mining algorithms, plus routines for data input and manipulation.
- This includes a variety of tasks such as pretty-print of decision trees, attribute subset, bagging and boosting, and alike. Orange also includes a set of graphical widgets that use methods from core library and Orange modules.
- Through visual programming, widgets can be assembled together into an application by a visual programming tool called **Orange Canvas**.
- All these together make the Orange tool, a comprehensive, component-based framework for machine learning and data mining, intended for both experienced users and researchers in machine learning who want to develop and test their own algorithms while reusing as much of the code as possible, and for those just entering who can enjoy in powerful while easy-to-use visual programming environment

KNIME

- KNIME (Konstanz Information Miner) is a user-friendly and comprehensive open-source data integration, processing, analysis, and exploration platform.
- From day one, KNIME has been developed using rigorous software engineering practices and is currently being used actively by over 6,000 professionals all over the world, in both industry and academia.
- KNIME is a modular data exploration platform that enables the user to visually create data flows (often referred to as pipelines), selectively execute some or all analysis steps, and later investigate the results through interactive views on data and models

Enterprise Miner (SAS)

- The Enterprise Miner's main tool for data visualization and descriptive statistics is the Insight module, which is a SAS system component. Insight offers a rich selection of tools for initial data analysis before the more complex modeling activities.
- Insight can be easily added to the analysis as one of the nodes in the Enterprise Miner network and, thus, it can accept not only raw data as input, but also data from data transformation modules (e.g., Replacement and Transform Variables).
- Insight can be run either on the entire dataset or a random sample.
- In its tutorial, SAS recommends against loading very large datasets into Insight and suggests that a sample of 2,000 observations is sufficient for most purposes
- The data visualization capabilities of Enterprise Miner are impressive, but they could still be significantly better if the tools to manipulate the characteristics of the charts and plots were easier to use.

Microsoft SQL Server

Microsoft SQL Server is a relational database server, developed by Microsoft: it is a software product whose primary function is to store and retrieve data as requested by other software applications, be it those on the same computer or those running on another computer across a network (including the Internet).

- Microsoft has introduced a wealth of new data mining features in Microsoft SQL Server 2008 that allow businesses to answer their concerns with data and mining for information in them.
- The current version of SQL Server, SQL Server 2008, (code-named "Katmai") aims to make data management self-tuning, self organizing, and self maintaining.
- SQL Server 2008 data mining features are integrated across all the SQL Server products, including SQL Server, SQL Server Integration Services, and Analysis Services.
- Accessing the data mining results is as simple as using an SQL-like language called Data Mining Extensions to SQL, or DMX.

Oracle

The **Oracle Database** (commonly referred to as *Oracle RDBMS* or simply as *Oracle*) is an object-relational database management system (ORDBMS) produced and marketed by Oracle Corporation.

- **Oracle Data Mining (ODM)** is used to incorporate data mining with the Oracle database.
- ODM is used for both supervised (where a particular target value should be specified) and unsupervised (where patterns in data are observed) data mining.
- The results of Oracle Data Mining can be viewed by the Oracle Business Intelligence's reporting/publishing component.
- Oracle BI Standard Edition One is a product that is used to extract business information concealed in the data.

- **Oracle Warehouse Builder (OWB)** is used to create the logical and physical design of the data mart.
- The **Oracle BI Server** is used to build a repository of metadata from the data mart that was created using the Oracle Data warehouse builder.
- The users ultimately interact with Oracle BI Answers to extract useful information from the data mart created using OWB (Oracle Warehouse Builder).
- **Oracle BI Interactive Dashboards** are used to publish the data extracted from the data mart so that the users can have an easy access to it.
- **Oracle BI publisher** is used to create reports that are very essential to any kind of business.

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ORACLE

ORACLE9I/I WAREHOUSE BUILDER: DIMENSIONAL DESIGN

Project Edit Object View Tools Window Help

ORACLE

- MY_PROJECT
- OWB904_DEMO
  - Collections
  - Databases
    - Oracle
      - HR
      - OE
      - STG
      - WH
        - Mappings
        - Transformations
        - Dimensions
          - CHANNELS
          - PRODUCTS
          - PROMOTIONS
          - SALES_REPS
          - T_TIME
          - WAREHOUSES
        - Cubes
          - INVENTORIES
          - SALES
        - Tables

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SPSS

SPSS (originally, Statistical Package for the Social Sciences) is a computer program used for survey authoring and deployment (**IBM SPSS Data Collection**), data mining (**IBM SPSS Modeler**), text analytics, statistical analysis, and collaboration and deployment (batch and automated scoring services).

Application of Data mining

- Data mining was originally developed to act as expert systems to solve problems and did not require assumptions to be made about data.
- Now it has little interest in the mechanics of the technique but requires understanding of data and business problem to predict the trends and behaviors in an automated way.
- However, data mining has become useful to sort out previously unknown patterns in a large data system, though often used for data dredging and data snooping - refers to the practice of misusing data mining techniques to show misleading or false conclusions

Application of Data mining

- Broadly, the data mining could be useful to answer the queries on –
- Forecasting with regard to what may happen in the future.
- Classifying things into groups by recognizing patterns.
- Associating similar events that are likely to occur together.
- Clustering the peoples into groups based on their attributes.
- Making the sequence what events are likely to lead to whom.

Fraud or non-compliance anomaly detection

- Data mining isolates the factors that lead to fraud, waste and abuse.
- The process of compliance monitoring for anomaly detection (CMAD)involves a primary monitoring system comparing some predetermined conditions of acceptance with the actual data or event.
- If any variance is detected (an anomaly) by the primary monitoring system then an exception report or alert is produced, identifying the specific variance.
- For instance credit card fraud detection monitoring, privacy compliance monitoring, and target auditing or investigative efforts can be done more effectively.

Intrusion detection

- It is a passive approach to security as it monitors information systems and raises alarms when security violations are detected.
- This process monitors and analyzes the events occurring in a computer system in order to detect signs of security problems. Intrusion detection systems (IDSs) may be either host based or network based, according to the kind of input information they analyze.
- Over the last few years, increasing number of research projects (MADAM-ID, ADAM, Clustering project, etc.) have been applied data mining approaches (either host based or network based) to various problems (construction of operational IDSs, clustering audit log records, etc.) of intrusion detection.

Lie detection (SAS Text Miner)

- SAS institute introduced lie-detecting software, called SAS Text Miner.
- Using intelligence of this tool, managers can be able to detect automatically when email or web information contains lies.
- Here data mining can be applied successfully to identify uncertainty in a deal or angry customers and also have many other potential applications.
- Many other market mining tools are also available in real practice viz. Clementine, IBM's Intelligent Miner, SGI's Mine Set, SAS's Enterprise Miner, but all pretty much the same set of tools.

Market basket analysis (MBA)

- Basically it applies data mining technique in understanding what items are likely to be purchased together according to association rules, primarily with the aim of identifying cross-selling opportunities.
- Sometimes it is also referred to as product affinity analysis.
- MBA gives clues as to what a customer might have bought if an idea had occurred to them.
- So, it can be used in deciding the location and promotion of goods by means of combo-package and also can be applied to the areas like analysis of telephone calling patterns, identification of fraudulent medical insurance claims,etc.

Aid to marketing or retailing

- Data mining could help direct marketers by providing useful and accurate trends on purchasing behavior of their customers and also help them in predicting which products their customers may be interested in buying.
- In addition, trends explored by data mining help retail-store managers to arrange shelves, stock certain items, or provide a certain discount that will attract their customers.
- In fact data mining allows companies to identify their best customers, attract customers, aware customers via mail marketing, and maximize profitability by means of identifying profitable customers.

Customer segmentation and targeted marketing

- Data mining can be used in grouping or clustering customers based on the behaviors (like payment history, etc.), which in turn helps in customer relationship management(epiphany)and performs targeted marketing.
- Usually it becomes useful to define similar customers in a cluster, holding on good customers, weeding out bad customers, identify likely responders for business promotions.

Phenomena of “beer and baby diapers”

- This story of using data mining to find a relation between beer and diapers is told, retold and added to like any other legend.
- The explanation goes that when fathers are sent out on an errand to buy diapers, they often purchase a six-pack of their favorite beer as a reward.
- An article in The Financial Times of London (Feb. 7, 1996) stated, "The oft-quoted example of what data mining can achieve is the case of a large US supermarket chain which discovered a strong association for many customers between a brand of babies nappies (diapers) and a brand of beer.

Financial, banking and credit or risk scoring

- Data mining can assist financial institutions in various ways, such as credit reporting, credit rating, loan or credit card approval by predicting good customers, risk on sanctioning loan, mode of service delivery and customer retention (i.e. build profiles of customers likely to use which services), and many others.
- A credit card company can leverage its vast warehouse of customer transaction data to identify customers most likely to be interested in a new credit product.
- In addition, data mining can also assist credit card issuers in detecting potentially fraudulent credit card transaction.
- In general, data mining methods such as neural networks and decision trees can be a useful addition to the techniques available to the financial analyst.

Medicare and health care

- Applying data mining techniques, it is possible to find relationship between diseases, effectiveness of treatments, to identify new drugs, market activities in drug delivery services, etc.
- However, a pharmaceutical company can analyze its recent sales to improve targeting of high-value physicians and determine which marketing activities will have the greatest impact in the next few months.
- The data needs to include competitor market activity as well as information about the local health care systems.
- Such dynamic analysis of the data warehouse allows best practices from throughout the organization to be applied in specific sale
- situation.

Corporate surveillance

- Implies the monitoring of a person or group's behavior by a corporation, which is highly possible through data mining process.
- So it can be used as a form of business intelligence that enables the corporation to better tailor their products and services to be desirable by their customers.
- Normally the organizations that have enemies who wish to gather information about the group members or activities face the issue of infiltration usually followed a process of data mining.
- Thus, surveilling party may put pressure on certain members of the target organization to act as informants i.e. disclose the information they hold on the organization and its members.

Scientific enquiry and research analysis

- Data mining can be applied in diverse areas of scientific activities and allows researchers to perform and analyze scientific data more effectively.
- It flourishes in telecommunication industry due to adequate amount of data available in structured form, which helps in network analysis, tracking unusual groups of customers, launching profitable service vouchers, etc.
- For instance, “if a customer makes more than X calls per hour to country Y, then apply treatment Z” –thus data mining (techniques) can be applied to all customers based on these (X, Y, Z) parameters.
- Many scientific enquiries in atmospheric science, biomedical engineering, information retrieval, patent analysis, remote sensing, etc. prefer to use data mining for novel discovery.

Criminal investigation and homeland security

- Data mining becomes useful in suspecting criminals as well as apprehending these criminals by examining trends in location, crime type, habit, and other patterns of behaviors.
- It also very useful to track fraud in order to increase the safety and security of a country.

Thank you !!!